

18. FORECASTING ASSEMBLY LINE DOWNTIME (REGRESSION)

Batch Members

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PROBLEM:

PROBLEM STATEMENT:

"FORECAST TOTAL DOWNTIME IN MINUTES PER DAY USING HISTORICAL AND OPERATIONAL DATA FROM AN ASSEMBLY LINE."

GOAL:

PREDICT DAILY DOWNTIME IN MINUTES.

IMPORTANCE:

ENABLES PROACTIVE MAINTENANCE

REDUCES PRODUCTION HALTS

IMPROVES EQUIPMENT EFFECTIVENESS AND OPERATIONAL EFFICIENCY

Dataset (Features)

Target Variable: Downtime_minutes

Features Used:

- Machine age
- Operator shift
- Previous downtime
- Number of breakdowns
- Task complexity
- Material quality
- Part rejection rate
- Equipment usage rate
- Safety incidents
- Line speed

<https://in.docworkspace.com/d/sIDGDh8Al3-LDwAY>

ALGORITHMS USED

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor
5. Support Vector Regressor (SVR)
6. XGBoost Regressor

Evaluation metrics used:

- R^2 Score
- RMSE

MODEL COMPARSION

✓ Model Comparison Results:

Model	R ² Score	RMSE	Verdict
Linear Regression	0.931	9.95	🏆 Best model
Gradient Boosting	0.872	13.58	Good, but not best
Random Forest	0.791	17.33	Fair performance
Support Vector	0.628	23.12	Weak performance
Decision Tree	0.564	25.05	Poor performance

CHOOSING LINEAR REGRESSION AS YOUR MODEL FOR FORECASTING ASSEMBLY LINE DOWNTIME OFFERS SEVERAL PRACTICAL BENEFITS:

✓ 1. HIGH ACCURACY IN THIS CASE

- IT ACHIEVED THE HIGHEST R^2 SCORE (0.931) AND LOWEST RMSE (9.95 MINS) AMONG ALL TESTED MODELS.
- THIS MEANS IT EXPLAINS OVER 93% OF THE VARIATION IN DOWNTIME — EXCELLENT PREDICTIVE POWER.

✓ 2. SIMPLICITY & INTERPRETABILITY

- LINEAR REGRESSION IS EASY TO UNDERSTAND AND EXPLAIN.
- YOU CAN SEE HOW EACH FEATURE (LIKE MACHINE AGE, BREAKDOWNS, ETC.) IMPACTS DOWNTIME, WHICH IS GREAT FOR DECISION-MAKING IN INDUSTRIAL SETTINGS.

✓ 3. LOW COMPUTATIONAL COST

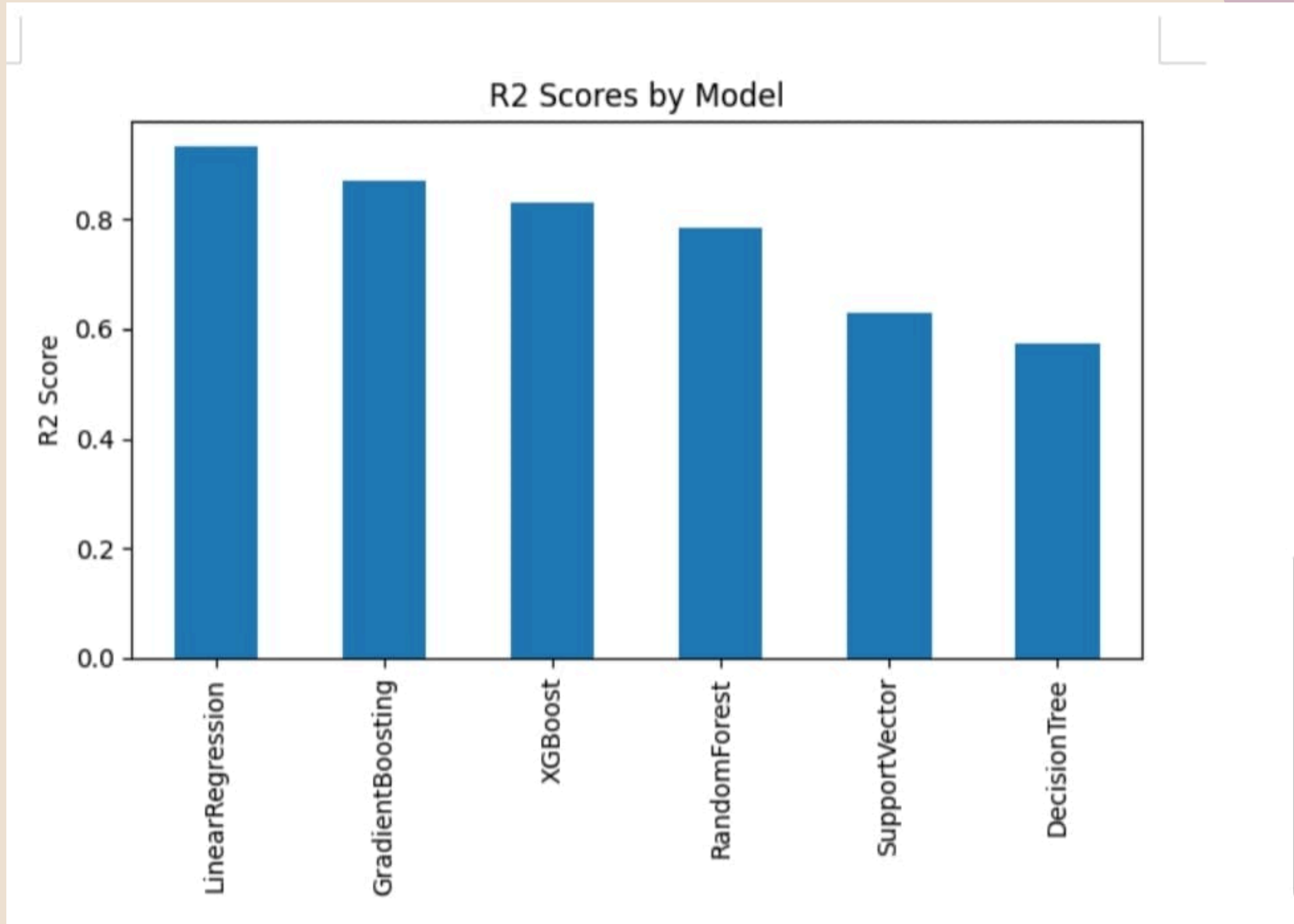
- IT TRAINS VERY FAST, EVEN WITH LARGE DATASETS.
- IDEAL FOR REAL-TIME OR FREQUENT RETRAINING SCENARIOS (LIKE DAILY UPDATES).

✓ 4. NO OVERFITTING (IN THIS CASE)

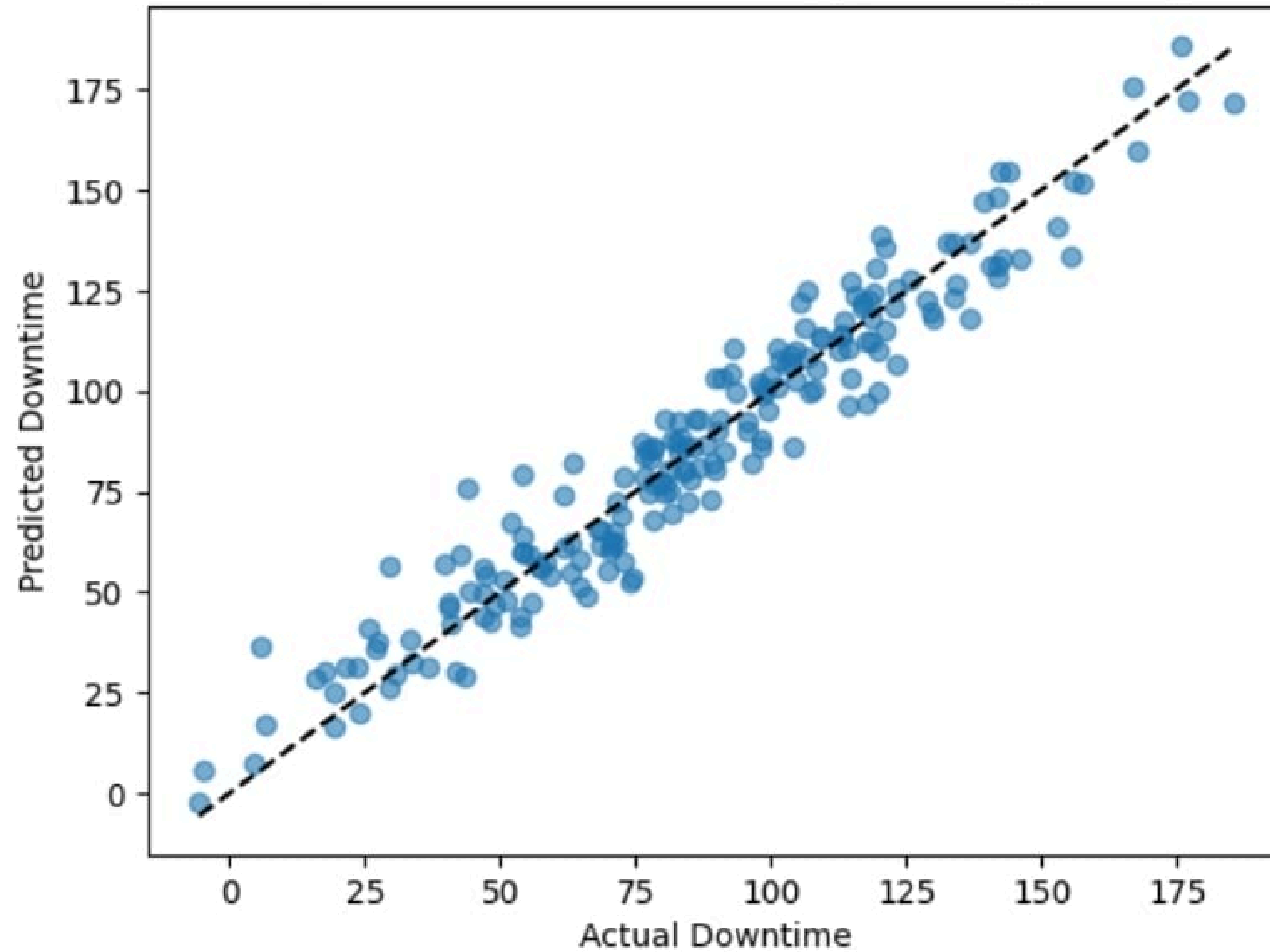
- UNLIKE COMPLEX MODELS (LIKE DECISION TREES OR SVR), LINEAR REGRESSION IS LESS PRONE TO OVERFITTING IF ASSUMPTIONS ARE MET.
- WITH PROPER SCALING AND LINEAR RELATIONSHIPS IN THE DATA (AS HERE), IT'S VERY EFFECTIVE.

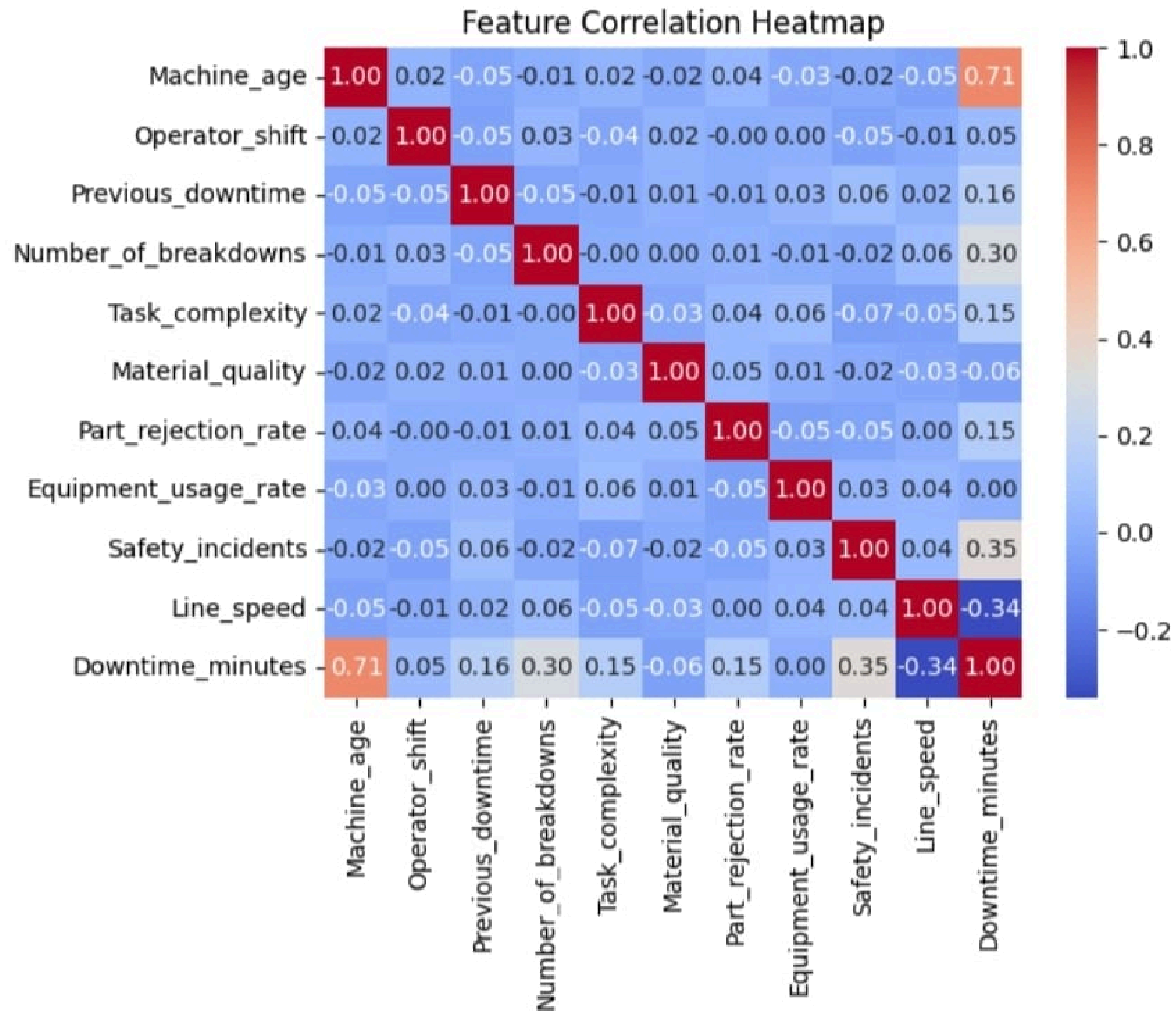
✓ 5. EASE OF DEPLOYMENT

- LINEAR REGRESSION MODELS ARE LIGHTWEIGHT AND EASY TO DEPLOY IN WEB APPS, EXCEL MACROS, AUTOMATION PIPELINES (E.G., IN A FACTORY DASHBOARD).



Actual vs Predicted Downtime (LinearRegression)





Q&A

1. WHAT WAS THE BEST-PERFORMING MODEL AND WHY?

LINEAR REGRESSION ($R^2 = 0.931$, RMSE = 9.95) DUE TO STRONG LINEAR RELATIONSHIPS AND EASY INTERPRETABILITY.

2. WHAT ARE R^2 AND RMSE, AND WHY ARE THEY USED?

R^2 SHOWS VARIANCE EXPLAINED; RMSE SHOWS AVERAGE PREDICTION ERROR. BOTH HELP EVALUATE MODEL PERFORMANCE.

3. HOW DID YOU VALIDATE THE MODEL?

USED 80/20 TRAIN-TEST SPLIT TO EVALUATE GENERALIZATION

4. WHY IS FEATURE SCALING IMPORTANT?

ENSURES FAIR CONTRIBUTION FROM ALL FEATURES, ESPECIALLY FOR MODELS LIKE SVR, KNN, AND LINEAR REGRESSION.

5. HOW IS THIS MODEL USEFUL IN REAL-WORLD MANUFACTURING?

ENABLES PROACTIVE MAINTENANCE, MINIMIZES DOWNTIME, AND IMPROVES PRODUCTIVITY PLANNING.

Thank you!!